A Reflective Journey

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I came to Iowa State not knowing much about my future. I enrolled in the Computer Engineering undergraduate program in 2024 without even researching what it entailed. My friend Jon had been planning a career in software engineering for years at the point of my decision, and I thought to myself, “I can’t do the exact same thing as him! I have to choose something different!” At that time, I had no idea how much the computer engineering program would shape me into an enthusiastic learner and problem solver, allow me to develop a deep interest into computer architecture, and positively influence my outlook on life.

Computer engineering has introduced me to all aspects of real-world problem solving. As of now, in my senior year, I am dedicating much of my time to my senior design project, an open-source car head unit. In our personal lives, Jon and I noticed a lack of solutions in one particular interest of ours: the car audio space. Specifically, one that satisfied the interests of a large set of hobbyists. Specific issues like high quality sound and sound processing, usability, modularity, portability, and cost were inhibiting many motivated users, a community that existing commercial products had priced out or ignored. We noticed their ability to find a head unit that suited their needs was inhibited by factors of economics and global production. In order to solve this problem, we applied our knowledge from embedded systems, software engineering and security, and ethics coursework to gather research on the communities’ desires while making sure to stay within our rights and capabilities. We spent weeks researching the required IEEE standards, hardware configurations, software stacks, and built out a detailed schema and testing plan to make sure we developed a useful solution.

Beyond coursework, I sought out my own resources to push this project forward: I utilized my research skills developed during my time at Iowa State to find out more about concepts and implementations of subjects like high quality Bluetooth audio codecs via Iowa State library resources and verified research papers from global scholars. I also discovered that my learning ventured beyond the classroom. I joined Iowa State’s ChipForge club, where I have learned about application-specific integrated circuit design in more depth than I ever could have in my coursework. I learned new

hardware description languages and led my group to design a full-fledged digital fast-Fourier transform circuit with real-world integration, which showed me that meaningful engineering growth can happen even more outside the classroom than in it.

My senior design project has also allowed me to grow in ways I didn't expect. One challenge that tested my ability to adapt quickly was taking charge of my chosen role to gather and research the requirements that members of the car audio community have voiced for years. Taking charge of an extremely important part of our project, that being the major components we needed to include to appeal to our audience, allowed me to experience taking charge of an engineering decision with real-world implications. We also faced a real ethical question when we considered implementing the Apple CarPlay protocol, giving us issues such as: "The community needs this, but we may not be within our full right to reverse engineer this feature". We had to dive deep into exploring all our options, communicating with professionals and experienced engineers, and giving time and thought to our actions to come up with the most responsible, economical solution that gave sufficient benefit to our users while abiding to our ethical standards and respecting all intellectual property laws. This experience taught me how to push past the technical components of my education to apply the ethical engineering experience I've gained. Through this experience, I realized that responsible engineering requires slowing down even when a deadline is pushing you to move fast. Moments like these changed my approach to learning. In my early years, I passively absorbed lecture material, cruised through my classes, and applied my effort to meeting the A-level requirements. I have since dedicated most of my time to applying my knowledge, seeking out experts and outside sources to further my knowledge, and exceeding the expectations of my professors and myself, the steps I needed to take to become an exceptional engineer.

Looking ahead, I want to continue the foundation I've built at Iowa State by applying my skills to an embedded systems or hardware design internship, computer engineering Master's degree, and an impactful career in a fulfilling field like agriculture technology or field-specific embedded systems engineering. My time at Iowa State has taught me how to keep learning, how to weigh responsibility against ambition and ambition against realism, and how to discover and create new solutions to real-world problems. That mindset will be the most valuable thing I take with me into the next stage of my life.